

Continuous spatial public goods game with self and peer punishment based on particle swarm optimization

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ABSTRACT

How cooperative behavior emerges and evolves in human society remains a puzzle. It has been observed that the sense of guilt rooted from free-riding and the sense of justice for punishing the free-riders are prevalent in the real world. Inspired by this observation, two punishment mechanisms have been introduced in the spatial public goods game which are called self-punishment and peer punishment respectively in this paper. In each situation, we have introduced a corresponding parameter to describe the level of individual tolerance or social tolerance. For each individual, whether to punish others or whether it will be punished by others depends on the corresponding tolerance parameter. We focus on the effects of the two kinds of tolerance parameters on the cooperation of the population. The particle swarm optimization (PSO)-based learning rule is used to describe the strategy updating process of individuals. We consider both of the memory and the imitation in our model. Via simulation experiments, we find that both of the two punishment mechanisms could facilitate the promotion of cooperation to a large extent. For the self-punishment and for most parameters in the peer punishment, the smaller the tolerance parameter, the more conducive it is to promote cooperation. These results can help us to better understand the prevailing phenomenon of cooperation in the real world.

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1. Introduction

The phenomenon of cooperation in human society and biological populations is a miracle. The explanation of the emergence and maintenance of cooperation in accordance with Darwin's theory of evolution remains a puzzle for academia [1,2]. The plight of cooperation can be well described by the model of prisoners' dilemma game with two people interacting or the public goods game (PGG) with many people participating. Theoretical scholars from multiple disciplines have tried to interpret the cooperation dilemma from different perspectives. Many cooperation mechanisms have been proposed, namely the punishment [3–7], the reward [8–10], the reputation [11–13], the optional participation [14–16], the population structure [17,18], and the recently proposed mechanisms of insurance compensation [19] and communication structure [20], and so on. Nowak has summarized five main mechanisms for the evolution of cooperation, including kin selection, direct reciprocity, indirect reciprocity, network reciprocity, and group selection [21].

We focus on the punishment and network reciprocity mechanisms in this paper.

Punishment, as a viable way to change human's behavior, is regarded as an important mechanism to maintain cooperation in human society. By behavioral experiments, experimental economists have found that punishment can significantly enhance the level of cooperation in the PGG [22]. Different forms of punishments have attracted the broad interest of scholars and various forms of costly punishments have been studied in the evolutionary PGG [23–29]. For example, the introduction of punishing strategies of punishing cooperators and punishing defectors [23,24]; the introduction of the moralists who cooperate while punishing non-cooperative behavior and immoralists who defect while punishing other non-cooperative behaviors [25]; self-organized punishment that allows players to adapt their sanctioning efforts in dependence on the success of cooperation [26]; conditional punishment that bind the fine on the number of other punishers [27]; probabilistic sharing of the responsibility to punish defectors [28]; and so on.

Network reciprocity mechanism considering the interactive structure of the population has also attracted widespread interest of researchers. Evolutionary games on networks are also termed spatial evolutionary games, in which the relationship between

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individual's micro-behavior and the system's macro-phenomenon need to be investigated, just like the propagation dynamics model in complex systems [30]. There has been a large amount of literature on the evolution of cooperation in spatial PGG (SPGG). Such as literature on different network structures [31–36], using different strategy update rules [37,38], and on dynamic networks which allows individuals to adjust their interactive objects when their strategies are updated [39–41]. Other literature focus on both network structure and other mechanisms, such as punishment [42–45], reputation [46,47], optional participation [48,49], emotional influence [50–52] and social impact [53]; and so on. The research progress of spatial evolutionary games and the evolution of cooperation can be found in review articles of Szabó et al. [54] and Perc et al. [55–57].

Both mechanisms of the punishment and network reciprocity are analyzed in this paper. A recent study shows that the punishment may diminish the benefits of network reciprocity in social dilemma [58]. In our model, we have also investigated the effect of punishment in the network reciprocity and the conclusion is different. Different from their research, we consider two forms of punishments: self-punishment which stems from the sense of guilt when one takes a free ride of others, the phenomenon of which has already been verified by public good experiments [59, 60], and peer punishment which stems from the sense of justice. In both cases, the punishments can be adjusted by some tolerance parameters. The peer punishment we have discussed is comparable to some former studies [23–25,27,28], but there are also some differences. As strategies in the standard PGG are discrete, the corresponding peer punishment punishes other individuals deterministically [23–25] or by probability [27,28]. But in our study, individuals have continuous strategic space. For an individual A , whether to punish individual B in the group game depends on a social tolerance parameter. Only if B 's investment is small enough (compared with A , the different part exceeds the social tolerance parameter), then it will be accordingly punished by A . Therefore, the peer punishment in our paper is based on social tolerance and it has more flexibility. When the social tolerance is close to 1, each individual will not punish others. Social tolerance here can be used as a parameter to adjust whether to punish others for individuals. It can be seen as a continuous model of the traditional public goods game with and without punishment. In this paper, we focus on the effects of the two kinds of tolerance parameters on the cooperation of the population.

It is worth mentioning that individuals' strategy updating rules or learning mechanisms will also have significant impacts on the evolutionary equilibrium of the system. In evolutionary games with discrete strategies, many learning mechanisms have been proposed, such as the Fermi rule [23,24,26,36,43,44], the Moran process-based rule [61,62], the directional learning rule [63], the win-stay-lose-shift rule [64–67], the aspiration-based rule [68–72]. However, in the continuous strategy situation, few learning rules are proposed. We observe that in the continuous spatial evolutionary games such as the SPGG, the strategy updating process of individuals has many characteristics in common with particle's behavior in the particle swarm optimization (PSO) algorithm. Firstly, individuals and particles both have strategies. In the SPGG, the individual's strategy is its investment; in the PSO, the particle's strategy is its position. Secondly, individuals and particles both have fitness values which are determined by their strategies. Thirdly, both individuals in the SPGG and particles in the PSO adjust their strategies based on their fitness. Fourthly, both individuals in the SPGG and particles in the PSO adjust their strategies through learning. In the PSO, the particle can learn the best positional information in their history and the position of the most adaptable particle. Therefore, in the SPGG, it is natural to assume that individuals can learn their best historical investment and the

strategy of the most successful neighbors. Inspired by this observation, we have adopted the PSO-based individual learning rule in our model. In fact, the PSO has been introduced into the individual's strategy updating process in the evolutionary games [73,74]. We focus on the effects of memory and imitation in the PSO-based learning process on the cooperation of the population.

The remainder of the paper is organized as follows. In Section 2, we describe the model of the continuous SPGG with two types of punishment and the PSO-based learning for individual's strategy updating. The main simulation results for the two punishment situations are presented in Section 3. In Section 4, we give some model extensions. Finally, we summarize the conclusions in Section 5.

2. Model

In our model, individuals are located on the nodes of a network and playing games with their direct neighbors. The network structure defines the objects that each individual can interact with. The strategy of each individual is the amount of its investment. The investment is continuous. They can choose any amount that does not exceed the maximum value. At every stage, each individual needs to participate in multi-group of PGGs at the same time. One is the group that is centered on himself (a group of nodes that directly connected to him), and other groups are centered on each of his direct neighbors. We assume that the strategy of individuals in the multiple games at the same time remains the same.

We only discuss homogeneous networks on which all nodes have the same degrees in this paper. Let $g_i(t)$ denote the strategy of individual i at time t , then the payoff of individual i participating in the game that centered on himself is:

$$\pi_{ii}(t) = \pi_0 - g_i(t) + \sum_{j \in \Omega_i \cup i} \frac{rg_j(t)}{|\Omega_i| + 1} \quad (1)$$

where Ω_i is the neighbor set for individual i . π_0 is the total amount that each individual can invest in one game. We set $\pi_0 = 1$, $0 \leq g_i(t) \leq \pi_0$. r is the investment amplification factor, $1 < r < |\Omega_i| + 1$; $|\Omega_i|$ is the number of neighbors for individual i .

The payoff of individual i participating in the game that centered on his neighbor j is:

$$\pi_{ij}(t) = \pi_0 - g_i(t) + \sum_{k \in \Omega_j \cup j} \frac{rg_k(t)}{|\Omega_j| + 1} \quad (2)$$

The total payoff $\pi_i(t)$ of individual i at time t is the sum of payoffs from all the $|\Omega_i| + 1$ round games it participated at time t .

$$\pi_i(t) = \pi_{ii}(t) + \sum_{j \in \Omega_i} \pi_{ij}(t) \quad (3)$$

In the following, we consider two forms of punishment respectively.

2.1. Self-punishment

For the self-punishment, we assume that the individual will have a sense of shame when his contributions are smaller below a certain criteria compared with his peers. This negative emotion in PGG has been already verified by behavior experiments in both psychology and economics literature [59,60]. The bad feeling will reduce individual's fitness. We think this feeling of shame is derived from the mutual comparison of the amount of investment between individuals. Therefore, we assume that the fitness reduction is a function of the number of individuals whose investment

amount exceeds a certain amount (h_{IT}). Let $f_i(t)$ denote the fitness of individual i at time t . Then

$$f_i(t) = \pi_i(t) - \gamma \sum_{k=1}^{|\Omega_i|+1} n_{s_k} \quad (4)$$

where $\gamma \sum_{k=1}^{|\Omega_i|+1} n_{s_k}$ is the fitness reduction owing to the shamed feeling of the focus individual when his contribution is not enough; n_{s_k} is the number of individuals in the k -th group of individual i whose investment exceeds its h_{IT} . Here the total number of groups individual i participates is $|\Omega_i| + 1$, and we denote the k -th group as G_k . That is, $n_{s_k} = \text{card}\{j \in G_k | g_i < g_j - h_{IT}\}$. Parameter h_{IT} ($0 < h_{IT} \leq 1$) denotes the individual tolerance. Parameter γ is the punishment intensity of the bad feeling on individual's fitness and we call it the guilt intensity in the following.

2.2. Peer punishment

For the peer punishment, each individual will be punished by other partners in his groups when his contribution is smaller than a certain criteria compared with other individuals. The fitness of the punished individuals will be reduced. Simultaneously, the focus individual will also punish other partners in his groups whose investment being less than his amount is beyond a certain value. Because of the cost of punishment, the fitness of punishers will also be reduced. As a consequence, the fitness of individual i at time t can be calculated as:

$$f_i(t) = \pi_i(t) - \beta \sum_{k=1}^{|\Omega_i|+1} n_{d_k} - \alpha \sum_{k=1}^{|\Omega_i|+1} n_{p_k}, \quad (\alpha < \beta) \quad (5)$$

where $\beta \sum_{k=1}^{|\Omega_i|+1} n_{d_k}$ is the fitness reduction owing to the punishment by other individuals; $\alpha \sum_{k=1}^{|\Omega_i|+1} n_{p_k}$ is the fitness reduction owing to the cost of punishing other individuals; n_{d_k} is the number of individuals in the k -th group of individual i whose investment exceeds its h_{ST} ; n_{p_k} is the number of individuals in the k -th group of individual i whose investment is lower than its h_{ST} . Here the total number of groups individual i participates is $|\Omega_i| + 1$, and we denote the k -th group as G_k . That is, $n_{d_k} = \text{card}\{j \in G_k | g_i < g_j - h_{ST}\}$, $n_{p_k} = \text{card}\{j \in G_k | g_i > g_j + h_{ST}\}$. Parameter h_{ST} ($0 < h_{ST} \leq 1$) denotes the social tolerance. β is the punishment intensity, and α is the unit punishment cost.

2.3. PSO-based strategy updating rule

The PSO algorithm is an evolutionary computing technology which stems from the research on bird foraging behavior. The basic idea is to search the optimal solution through the cooperation and information sharing among individuals in the group. The search process can be described as flying a swarm of particles with each occupying a position representing a possible solution to the optimization problem, the function value of which will determine its fitness. In the search process, each particle has its own velocity and each can adjust to move closer to the best particle in its neighborhood, as well as toward its own best position obtained thus far to increase its fitness. Let $x_i(t)$, $v_i(t)$ denote the position and velocity of individual i at time t respectively, $gbest_i(t)$ denote the position of the best particle in i 's neighborhood at time t , and $pbest_i(t)$ denote i 's best position obtained thus far at time t . Then particles update their speed and position with the following formulas: $v_i(t+1) = wv_i(t) + c_1 \times \text{rand}() \times [pbest_i(t) - x_i(t)] + c_2 \times \text{rand}() \times [gbest_i(t) - x_i(t)]$ and $x_i(t+1) = x_i(t) + v_i(t)$, where parameter w denotes the inertia weight, the value of which can be utilized to adjust the particle's global search capability. c_1 and

c_2 are two learning parameters; $\text{rand}()$ is a random number between 0 and 1.

Similar to the behavior of particles in PSO, during the evolutionary process of the games, each individual adjusts its strategy based on its fitness in the last step. Inspired by the idea of the PSO algorithm, it is natural to assume that each individual is allowed to learn from the strategy in its local neighborhood (including the individual itself) that has the highest fitness in the last step; in addition, individual has memory that it can remember its best strategy in the past steps.

Based on the learning process described above, and adopting the specific learning parameters as in Ref. [73], individual i 's speed to adjust its strategy at time $t+1$ can be described as:

$$v_i(t+1) = v_i(t) + \omega[g_i^*(t) - g_i(t)] + (1 - \omega)[g_{U_i}^*(t) - g_i(t)] \quad (6)$$

where $g_i^*(t)$ denotes the history strategy of individual i at time t that has the highest fitness. That is:

$$g_i^*(t) = g_i(t^*), \quad t^* = \arg \max_{t_0 \leq t} f_i(t_0) \quad (7)$$

$g_{U_i}^*(t)$ is the most successful neighbor's strategy for individual i (including itself) that has the highest fitness at time t . That is:

$$g_{U_i}^*(t) = g_{j^*}(t), \quad j^* = \arg \max_{j \in \Omega_i \cup i} f_j(t) \quad (8)$$

Parameter ω ($0 \leq \omega \leq 1$) indicates the memory factor weight (or $1 - \omega$ the imitation factor weight).

At the initial time $t = 0$, each player is randomly assigned a strategy which is uniformly from the interval $[0, 1]$ and let $v_i(0) = 0$. Then the strategy of individual i at time $t+1$ can be described as:

$$g_i(t+1) = g_i(t) + v_i(t+1) \quad (9)$$

In the strategy updating process, we have the following boundary conditions. When $g_i(t+1) > 1$, let $g_i(t+1) = 1$. When $g_i(t+1) < 0$, let $g_i(t+1) = 0$. Each player updates its strategy synchronously.

3. Results and discussions

The simulations are carried out on the 100×100 square lattice with periodic boundary conditions. There are $N = 10000$ agents located on the vertices of the network. The average level of cooperation $\bar{g}$ is calculated by

$$\bar{g} = \frac{1}{N} \sum_{i=1}^N g_i(t) \quad (10)$$

The results are obtained through averaging over 10 independent runs. Each equilibrium cooperation frequency is calculated through averaging the last 500 generations after a transient period of 50000 generations.

3.1. Self-punishment

In order to understand the impact of individual tolerance h_{IT} on the average cooperation level of the population, parameters $\omega = 0.5$, $\gamma = 0.1$ are fixed to study the relationship between the cooperation level and r under different values of h_{IT} . Fig. 1 shows corresponding five comparison curves when h_{IT} equals 0.1, 0.3, 0.5, 0.7 and 1 respectively. It can be seen from the figure, the value of h_{IT} has a great impact on the cooperation level. When h_{IT} is equal to one, the individual has the highest tolerance, and for any

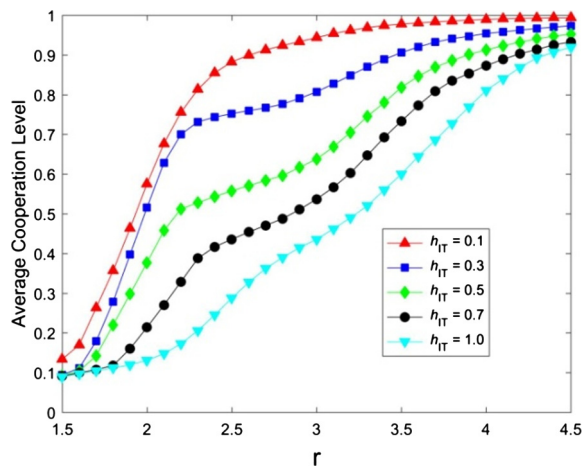


Fig. 1. In the self-punishment situation, comparison curves of the average level of cooperation in the population with r under different values of h_{IT} for fixed $\omega = 0.5$, $\gamma = 0.1$.

amount of investment, the individual does not feel ashamed which is equivalent to the case of the absence of self-punishment. The smaller the h_{IT} , the more likely the individual is to feel ashamed and the more likely it is to promote cooperation. The spatial distribution of strategies under four different values of h_{IT} for fixed $\omega = 0.5$, $\gamma = 0.1$, $r = 2.5$ at time step 50000 is shown in Fig. 2.

The effect of guilt intensity γ on the average cooperation level of the population can be seen in Fig. 3. And Fig. 4 indicates the

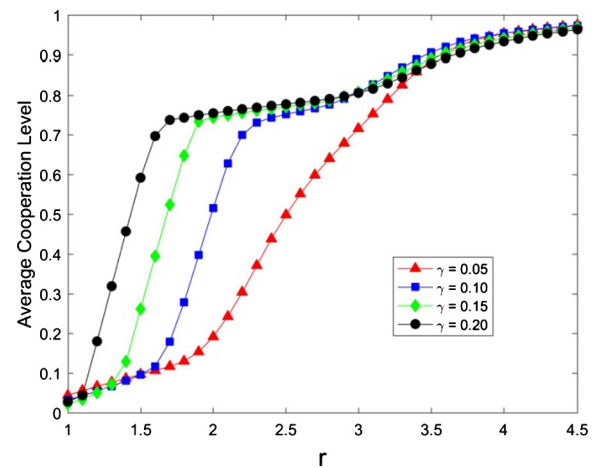


Fig. 3. In the self-punishment situation, comparison curves of the average level of cooperation in the population with r under different values of γ for fixed $\omega = 0.5$, $h_{IT} = 0.3$.

spatial distribution of strategies under the corresponding four values of γ for fixed $r = 2.0$ at time step 50000. Obviously, individual guilty emotion can promote cooperation to a large extent. Within a certain range of parameters, the greater the γ , the higher the level of cooperation in the population. As shown in Fig. 3, when γ is large ($\gamma = 0.2$), small values of r ($r = 1.5$) can facilitate the population to achieve a high level of cooperation.

Fig. 5 presents some comparison curves of the average level of cooperation in the population with r under five different values

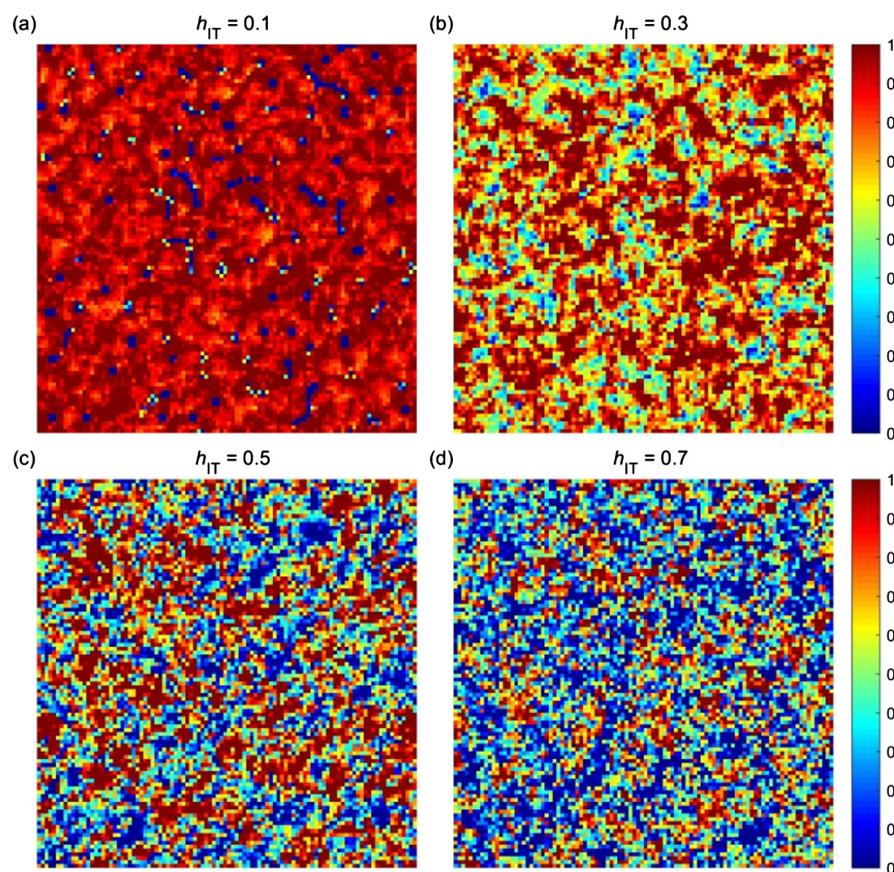


Fig. 2. In the self-punishment situation, spatial distribution of strategies under four different values of h_{IT} for fixed $\omega = 0.5$, $\gamma = 0.1$, $r = 2.5$. The snapshots are obtained at time step 50000. The color map encodes the cooperation level of each player. The dark red one represents for full cooperation ($g = 1$) and the dark blue one represents for full defection ($g = 0$). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

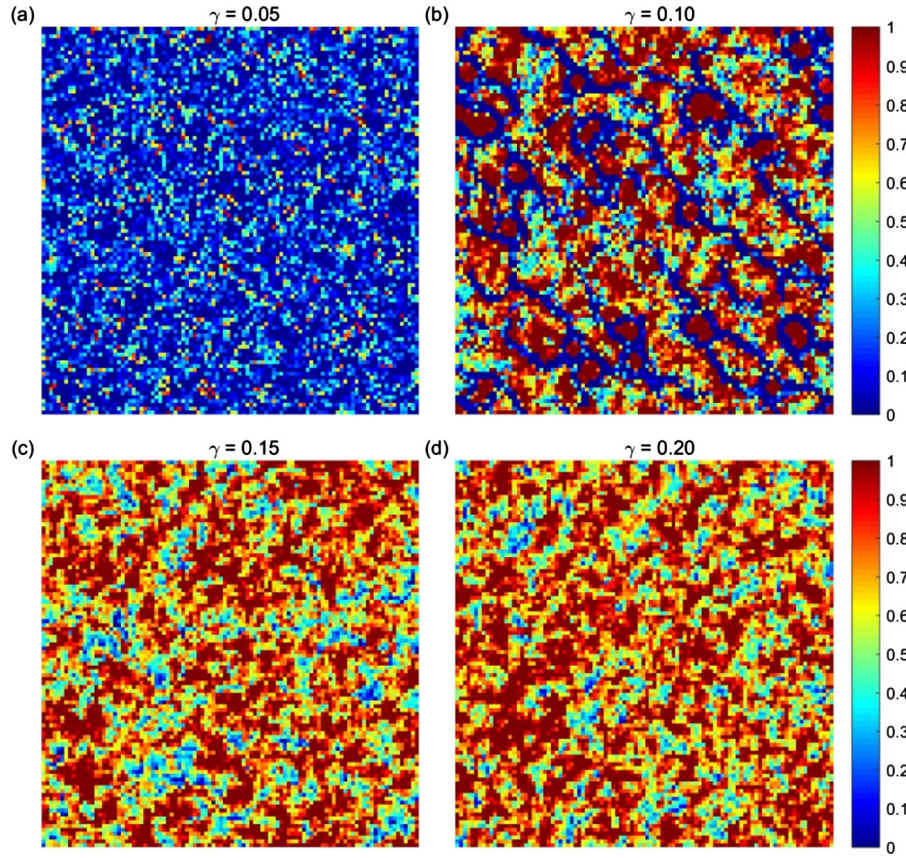


Fig. 4. In the self-punishment situation, spatial distribution of strategies under four different values of γ for fixed $\omega = 0.5$, $h_{IT} = 0.3$, $r = 2.0$. The snapshots are obtained at time step 50000.

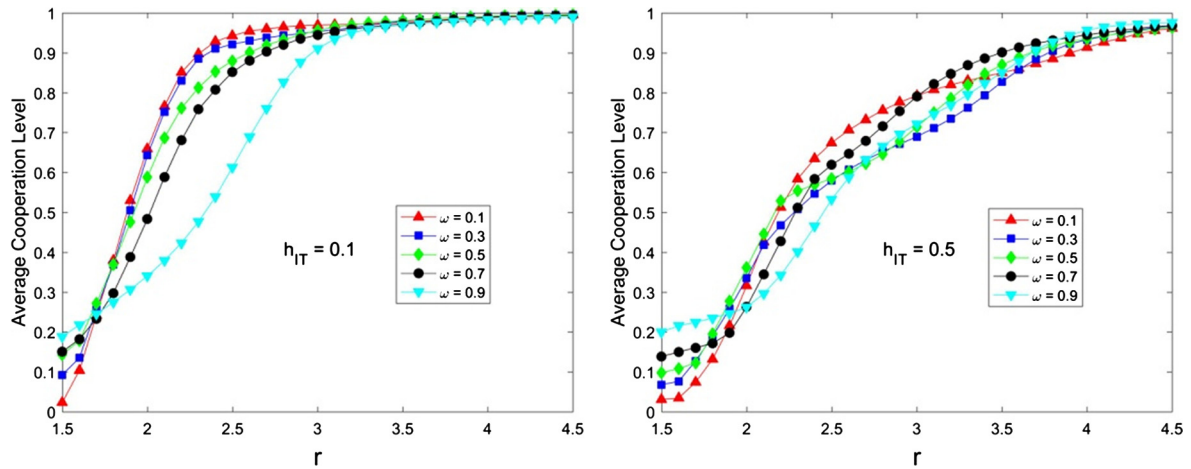


Fig. 5. In the self-punishment situation, comparison curves of the average level of cooperation in the population with r under different values of ω for fixed $\gamma = 0.1$ and $h_{ST} = 0.1, 0.5$ respectively.

of ω for fixed $\gamma = 0.1$ and two values of $h_{IT} = 0.1, 0.5$ respectively. As shown in the figure, there is a very wide range of parameters that can facilitate the population to achieve very high levels of cooperation in both of the two cases. The reduction of h_{IT} can expand the parameter ranges of achieving high levels of cooperation.

Fig. 6 illustrates the comparisons of the evolutionary process of the average level of cooperation under five different values of h_{IT} over time. The figure gives the evolution process within 10^5 time steps under the log scale.

3.2. Peer punishment

For the peer punishment, we first fix parameters $\omega = 0.5$, $\alpha = 0.1$, $\beta = 0.2$ to study the relationship between the cooperation level and r under different social tolerance h_{ST} . Fig. 7 shows five comparison curves when h_{ST} equals 0.1, 0.3, 0.5, 0.7 and 1 respectively. As shown in the figure, the effect of h_{ST} on the cooperation is somewhat similar to that of h_{IT} in the self-punishment. The peer punishment based on social tolerance can effectively promote the evolution of cooperation. For most parameter values, the

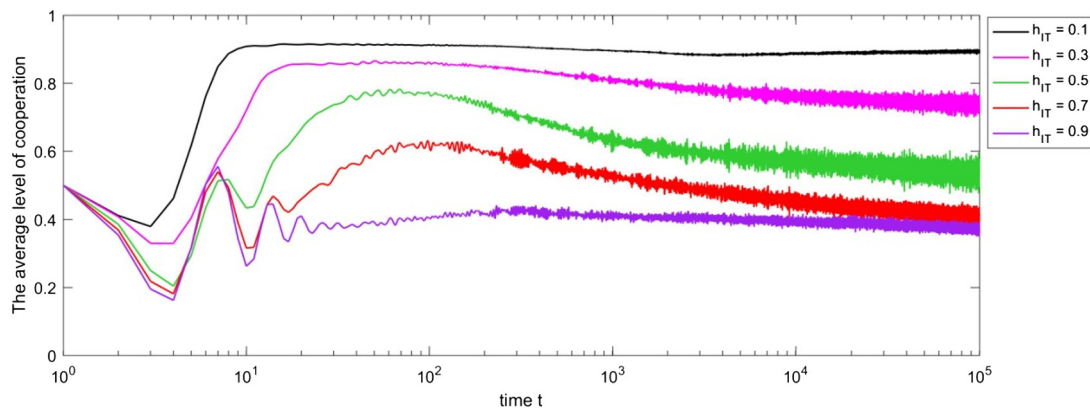


Fig. 6. In the self-punishment situation, comparisons of the average level of cooperation in the population over time under five different values of h_{IT} for fixed $\omega = 0.5$, $\gamma = 0.1$, $r = 2.5$.

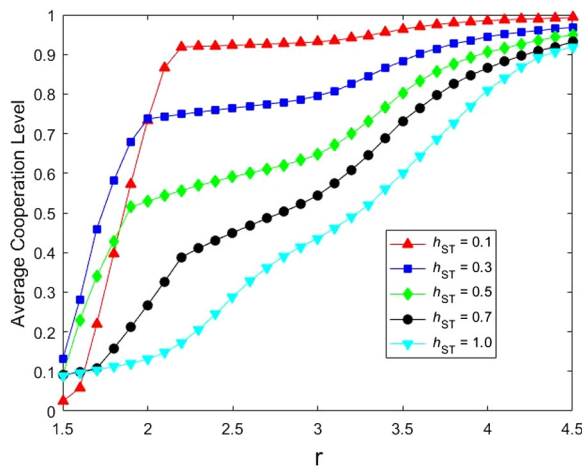


Fig. 7. In the peer punishment situation, comparison curves of the average level of cooperation in the population with r under different values of h_{ST} for fixed $\omega = 0.5$, $\alpha = 0.1$, $\beta = 0.2$.

smaller the h_{ST} , the more likely the individual is to be punished by its peers, and the more likely it is to promote cooperation. However, the conclusion no longer holds when the social tolerance and the investing amplification factor are both small. We can see the results corresponding to $h_{ST} = 0.1$ and $r < 2$. In this situation, a smaller h_{ST} cannot promote the cooperation to a higher level. We think it can be explained as follows. When r is very small, most individuals have low levels of investment. If the social tolerance is very low ($h_{ST} = 0.1$), for an individual, the cost of punishment brought about by increasing its investment will be greatly increased, and the smaller the h_{ST} , the more the punishment cost increases. Thus, the reduction of h_{ST} will have negative effects on cooperation. The spatial distribution of strategies under four different values of h_{ST} for fixed $\omega = 0.5$, $\alpha = 0.1$, $\beta = 0.2$, $r = 2.2$ at time step 50000 is shown in Fig. 8.

Fig. 9 reveals the contrast curves of the relationship between the cooperation level and r under four different values of β . Fig. 10 shows the spatial distribution of strategies under the four corresponding values of β for fixed $r = 1.7$ at time step 50000. From the two figures, the effect of punishment intensity β on the average level of cooperation can be noticed. Obviously, increasing the intensity of punishment can greatly facilitate the evolution of cooperation. Within a certain range of parameters, the greater the β , the higher the level of cooperation in the population. As shown in Fig. 9, when β is large ($\beta = 0.3$), small values of r ($r = 1.5$) can facilitate the population to achieve a high level of cooperation.

The effect of β is comparable to γ in the self-punishment situation.

Fig. 11 gives some comparison curves of the average level of cooperation in the population with r under five different level of ω for fixed $\alpha = 0.1$, $\beta = 0.2$ and two values of $h_{IT} = 0.1, 0.5$ respectively. As shown in the figure, there is a very wide range of parameters that can facilitate the population to achieve very high levels of cooperation in both cases. The most significant difference from the case of the self-punishment is that when r is not too small ($r \geq 2$), the reduction of h_{ST} can facilitate the promotion of cooperation. However, when r is small ($r < 2$) and h_{ST} is reduced to a certain value ($h_{ST} = 0.1$), the reduction of h_{ST} cannot promote any further cooperation.

Fig. 12 shows the comparisons of the evolutionary process of the average level of cooperation in the population under five different values of h_{ST} over time. The figure gives the evolution process within 10^5 times under the log scale.

4. Model extensions

4.1. Heterogeneous population

In the previous study, individuals are supposed to be homogeneous and all have same values of social tolerance parameter. But in reality, it looks not reasonable that everyone would like to punish others because of the cost. If some individuals do not punish others, it can be considered that their social tolerance values are equal to 1. We consider this situation with a heterogeneous population in this section. Taking the peer punishment case as an example, we assume that there are two types of individuals, one type with social tolerance $h_{ST} < 1$ and the other type with $h_{ST} = 1$ (not punishing others). These two types of individuals are evenly distributed in the network. Let θ denote the proportion of individuals with $h_{ST} = 1$. Fig. 13 shows five comparison curves of the average level of cooperation in the population varying with r under five different values of θ when other parameters are fixed.

4.2. Limited memory length

In the previous study, individuals are supposed to have infinite memory which can remember all their past fitness. In this section, we consider the limited memory length situation. Taking the peer punishment case as an example, we consider three memory lengths of 100, 300, 500 respectively, and compare them with the results of the infinite memory case. Fig. 14 shows four comparison curves of the average level of cooperation in the population

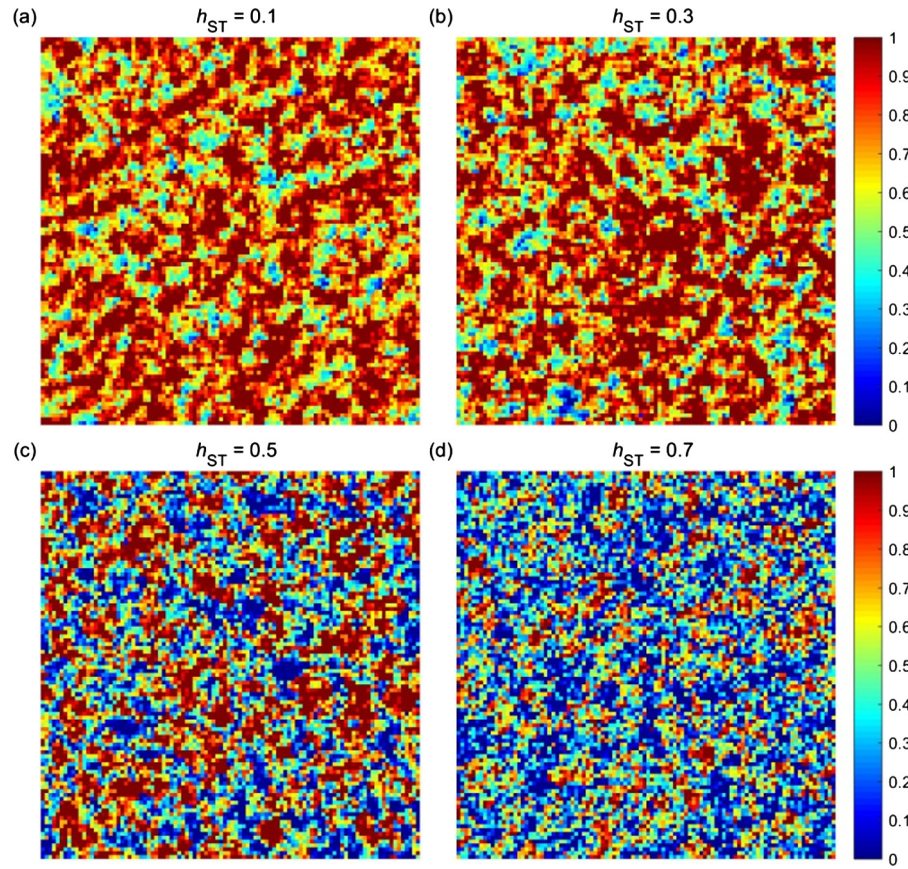


Fig. 8. In the peer punishment situation, spatial distribution of strategies under four different values of h_{ST} for fixed $\omega = 0.5$, $\alpha = 0.1$, $\beta = 0.2$, $r = 2.2$. The snapshots are obtained at time step 50000.

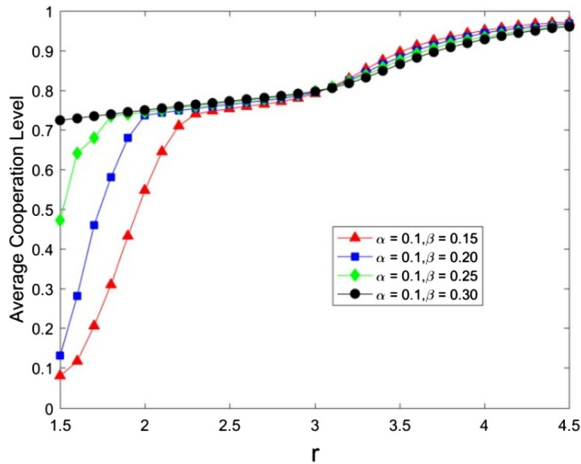


Fig. 9. In the peer punishment situation, comparison curves of the average level of cooperation in the population with r under different values of (α, β) for fixed $\omega = 0.5$, $h_{ST} = 0.3$.

varying with r under four different memory lengths when other parameters are fixed.

4.3. With mutation

In the previous study, we assume that individuals update their strategy based on formula (9) which ignores mutation. In this section, we consider the mutation situation. Taking the peer punishment case as an example, we introduce a mutation parameter ε , with which probability each individual's strategy will mutate

in their strategy updating process. We assume that the variation strategy obeys the uniform distribution on the interval of 0 to 1. Fig. 15 shows four comparison curves of the average level of cooperation in the population varying with r under four mutation rates when other parameters are fixed.

4.4. Asynchronous updating

In the previous study, we assume that individuals update their strategy synchronously. In this section, the asynchronous updating situation will be discussed. Taking the self-punishment case as an example, Fig. 16 shows comparisons between the synchronous and asynchronous updating rules under different values of h_{IT} for fixed $\omega = 0.5$, $\gamma = 0.1$.

5. Conclusions

In this paper, the evolutionary public goods game with continuous strategy space on the square lattice network has been elaborated. Two kinds of punishment mechanisms are introduced. One is self-punishment owing to guilty feeling of individuals, and the other is peer punishment due to social justice. For each individual, whether to punish others or whether it will be punished by others depends on a parameter of individual tolerance and social tolerance respectively. We focus on the effects of the two kinds of tolerances on the cooperation of the population. Under these two punishment mechanisms, we apply the PSO learning to the individual's strategy updating process. Both memory and imitation are considered in the model in which individuals can learn their best history strategies and best neighbors in

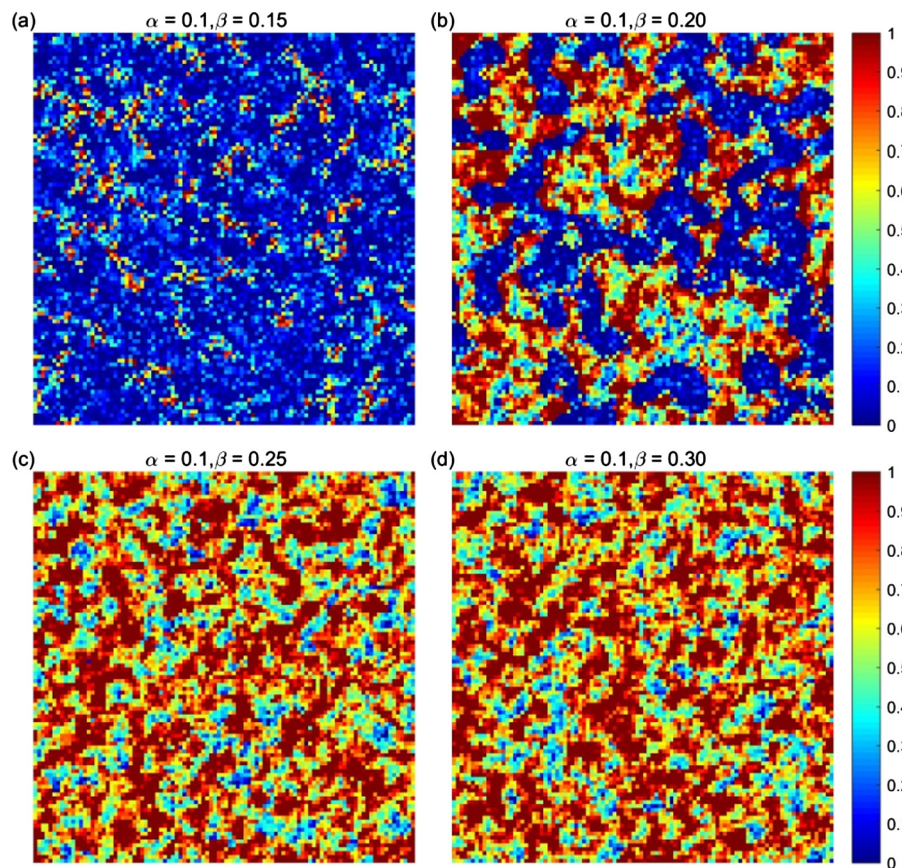


Fig. 10. In the peer punishment situation, spatial distribution of strategies under four different values of (α, β) for fixed $\omega = 0.5$, $h_{ST} = 0.3$, $r = 1.7$. The snapshots are obtained at time step 50000.

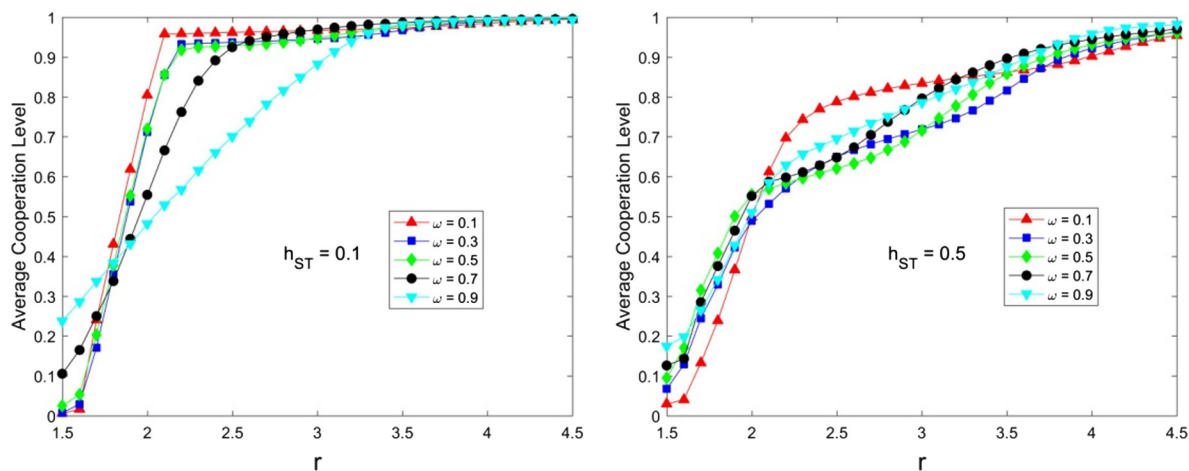


Fig. 11. In the peer punishment situation, comparison curves of the average level of cooperation in the population with r under different values of ω for fixed $\alpha = 0.1$, $\beta = 0.2$ and $h_{ST} = 0.1, 0.5$ respectively.

their neighborhood at every moment. The effects of memory and investment amplification factor on the cooperation are also investigated. By simulation experiments, we find that both of the self-punishment and peer punishment mechanisms could facilitate the promotion of cooperation significantly. In the self-punishment and for most parameters in the peer punishment, the smaller the tolerance parameter, the more conducive it is to promote cooperation. The influence of the memory parameter on the cooperation has two sides. All these conclusions can help to better un-

derstand the prevailing phenomenon of cooperation in the real world.

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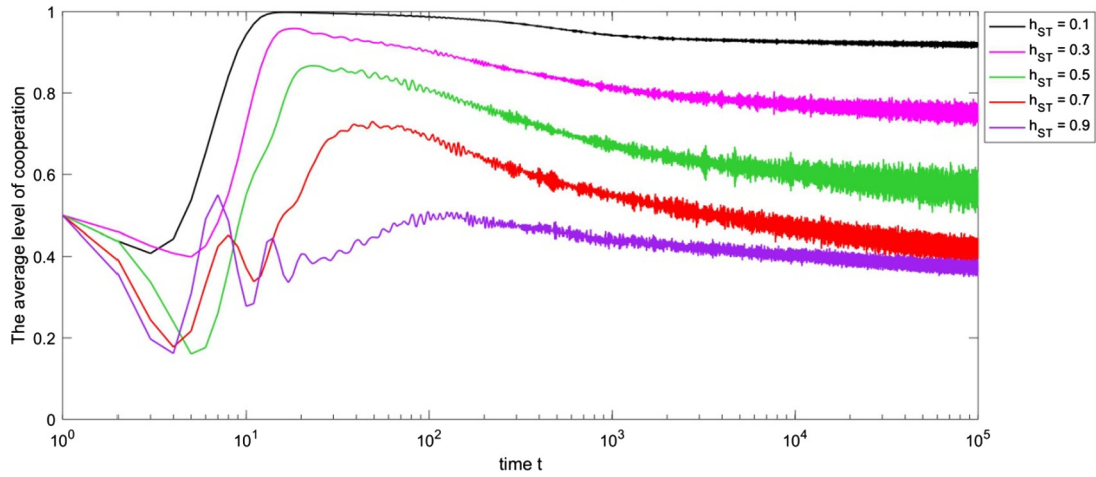


Fig. 12. In the peer punishment situation, comparisons of the average level of cooperation in the population over time under five different values of h_{ST} for fixed $\omega = 0.5$, $\alpha = 0.1$, $\beta = 0.2$, $r = 3$.

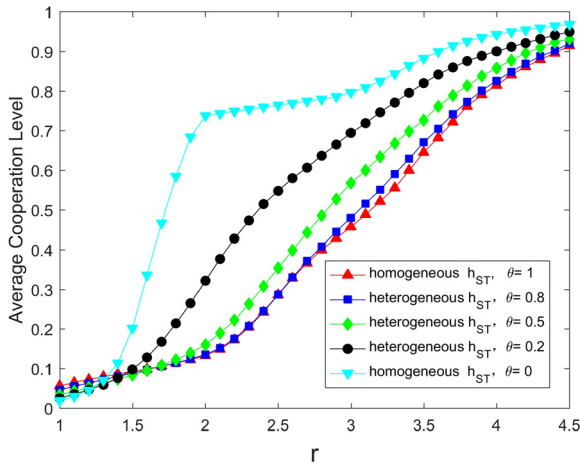


Fig. 13. In the peer punishment situation, comparison curves of the average level of cooperation in the population varying with r under different values of θ for fixed $h_{ST} = 0.3$, $\omega = 0.5$, $\alpha = 0.1$, $\beta = 0.2$.

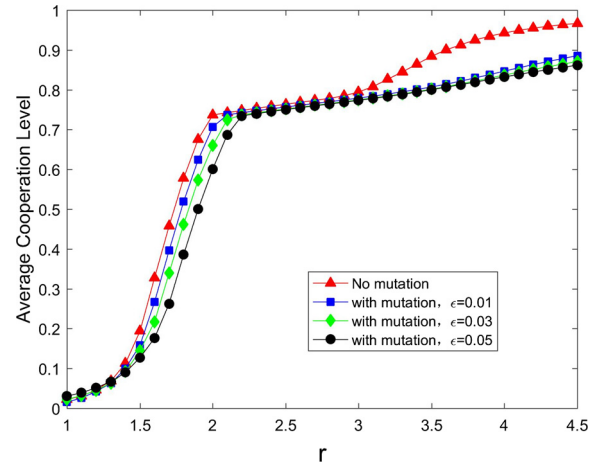


Fig. 15. In the peer punishment situation, comparison curves of the average level of cooperation in the population varying with r under different mutation rates for fixed $h_{ST} = 0.3$, $\omega = 0.5$, $\alpha = 0.1$, $\beta = 0.2$.

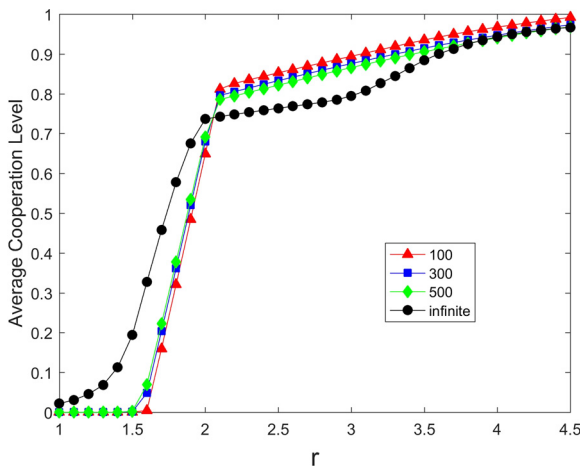


Fig. 14. In the peer punishment situation, comparison curves of the average level of cooperation in the population varying with r under different memory lengths for fixed $h_{ST} = 0.3$, $\omega = 0.5$, $\alpha = 0.1$, $\beta = 0.2$.

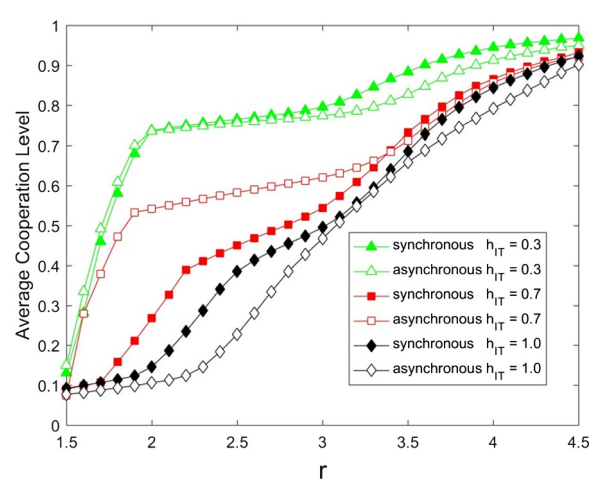


Fig. 16. In the self-punishment situation, comparisons between the synchronous and asynchronous updating rules under different values of h_{IT} for fixed $\omega = 0.5$, $\gamma = 0.1$.

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